AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM240
Introduction to computer graphics

Final Examination
Fall Semester 2025/2026

Date:14/Jan./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: This part consists of 10 multiple choice questions. For each question, encircle the label a, b, c, or d as correct answer. Each question carries 2 marks.
Part II: This part consists of four questions with a total of 80 Marks, solve all questions.

1 the junctions between individual Bezeier pieces in B-spline curve are called.

A modifiers B knots
C controllers D None of the above

2 The Java library .....contains advanced image processing and comprehensive image processing.

A AWT B Java2D
C Swing D JAI

3 The .....can be viewed as a part of color properties that specifies the transparency

A Alpha B Red
C Intensity D None of the above

4 The stroke of a line covers the attribute .....

A line width B color
C font D None of the above

5 The .....colour system is commonly used in printers.

A CIEXYZ B RGB
C CYMK D None of the above

6 the function ..... Is used to apply an operator on a given image.

A LookupOp B ConvolveOp
C filter D RescaleOp

7 the function .....can be used to create new color map for an image

A RescaleOp B ConvolveOp
C ColorConvertOp D LookupOp

8 in .....curves, only requires that the curve gets close to the control points, but it does not have to pass through them

A interpretation B Interpolation
C Approximation D None of the above

9 .....produces dynamic graphics contents and rendering.

A Viewing B Animation
C Interaction D None of the above

10 The top left corner of the ellipse defined by Ellipse2D.Float(20f,30f,100f,80f) is at .....

A (80,70) B (100,80)
C (20,80) D (20,30)

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Question 2 [20 Marks]
Write a java code segment to draw the shape:



Question 3 [20 Marks]

a. Explain the role and components involved in the animation process of a graphical application. [5 marks]

b. Having the following shape: [15 marks]

Write a java code segment to plot it assuming that the lines inside the font are randomly positioned.



Question 4 [20 Marks]

Having an image object of type BufferedImage that has the following artifacts:

1. Image is very dark
2. image is sheared towards the negative y-directions
3. image is rotated anti-clockwise around its center
4. image has inverted colors
5. image is zoomed in

Write a java code segment to resolve the above problems.

Question 5 [20 marks]

a. Explain how the dash style in line styling is defined and provide an example of its application. [5 marks]

b. Given the following list of control points:
(0,6)
(0,41)

a. Derive the parametric Bezier equation for the curve controlled by the above point list.
b. Write a code to implement the parametric equations that you derived in the previous point.

End of questions

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